

William Marcus

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Education

•Carleton University

Graduated April 2026

Bachelor of Computer Science Honours (Co-op)

Overall CGPA: 10.89/12 | Major GPA: 11.33/12

Experience

•Dayforce HCM

January 2025-August 2025

Software Engineer In Test Intern

Remote

- Developed and executed automated test scripts using the FitNesse framework, reducing manual testing by 90%.
- Designed and implemented comprehensive test cases and testing strategies, increasing automated code coverage and improving application reliability and release quality.
- Optimized existing automation suites, reducing execution time by over 50%, improving CI/CD pipeline efficiency.

•Bank of Canada

May 2024-August 2024

Full-stack Developer Intern

Ottawa, Ontario

- Developed and maintained features using PHP, JavaScript, HTML, and CSS across five major Bank of Canada websites, including BankofCanada.ca.
- Collaborated within an Agile development environment, participating in code reviews, daily scrum meetings, and sprint planning sessions.
- Performed comprehensive testing using PHPUnit and Behat to ensure application stability, code quality, and reliable deployments.

•Bank of Canada

May 2023-March 2024

Application Developer Intern

Ottawa, Ontario

- Migrated Jira application data to an upgraded platform using JavaScript, Node.js, Axios, and REST APIs, improving data accessibility and system maintainability.
- Developed responsive and user-friendly interfaces using React.js to enhance usability and user experience.
- Designed and implemented a ticket management tool using Node.js to organize and track service requests efficiently.

Personal Projects

•Pac-Man AI Reinforcement Learning Simulation

December 2025

<https://github.com/WillMarcuss/Pac-Man-RL-Project>

- **Tools & technologies used:** Python | NumPy | Pygame | Matplotlib | JSON-based Q-table serialization | Gymnasium- style custom environments | Linear Function Approximation | Neural Networks | PyTorch |
- **Description:** Developed a fully functional Pac-Man-style environment as part of a 4-member group project, featuring multiple maps, ghosts, pellets, and reward shaping. Implemented several RL agents (Dyna-Q, SARSA, DQN, PPO) and evaluated their performance across diverse map sizes and state spaces, achieving successful policy learning and environment-adaptive behaviors.

Technical Skills

Programming Languages: Java, Python, C, C++, JavaScript, TypeScript, HTML, CSS, SQL, PHP, C#, Swift, Haskell, Gherkin

Frameworks & Libraries: React.js, Angular, Node.js, .NET, WPF, Selenium, Cypress, FitNesse, PHPUnit, JUnit, Jest, Cucumber, Axios, Pygame, JavaFX, Java Swing, Java AWT

Software Development Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms, REST APIs, Test Automation, Agile Development, CI/CD, Software Testing, Reinforcement Learning

Tools & Technologies: Git, GitHub, PyTorch, NumPy, Matplotlib, JSON, SQL Databases